EXPLORATORY DATA ANALYSIS (EDA) REPORT

Submitted By:

Chukwuemeka Nnaji

Course:

Data Analytics

Title:

Exploratory Data Analysis (EDA) on Clean Dataset

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1. Introduction

Exploratory Data Analysis (EDA) is a process used to examine datasets to discover patterns, identify anomalies, test assumptions, and summarize characteristics using statistical and graphical methods.

The objective of this analysis is to explore the provided dataset to understand distributions, identify trends, detect outliers, and generate meaningful business insights.

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2. Objectives

The analysis was conducted to:

- Calculate descriptive statistics (mean, median, count)
- Identify patterns and trends within the dataset
- Detect outliers and unusual observations
- Generate insights from the available data
- Provide recommendations for further analysis

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3. Dataset Overview

The dataset contains transactional and customer-related information.

Dataset Summary

Metric	Value
Total Records	1,200
Total Variables	14
Data Status	Cleaned and ready for analysis

The variables include order information, customer activities, pricing details, payment methods, and sales performance indicators.

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4. Descriptive Statistics

The following descriptive statistics were calculated.

Variable	Count	Mean	Median
Quantity	1200	2.95	3
Unit Price	1200	356.41	364.21
Items in Cart	1200	5.48	5
Total Price	1200	1053.97	823.62

Interpretation

- Average customer purchase quantity is approximately **3 units per transaction**.
- Average selling price per unit is **356.41**.
- Customers purchase approximately **5 items per cart**.
- Mean Total Price is higher than the median, indicating a possible right-skew caused by larger purchases.

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5. Trend Analysis

Monthly sales patterns were examined.

Highest Revenue Months

Month	Revenue
June 2024	68,068.54
May 2023	63,836.84
January 2023	56,685.75
August 2023	54,352.14

Month	Revenue
June 2025	53,047.40

Findings

- June 2024 recorded the highest revenue.
- Sales fluctuate across months, indicating potential seasonality and demand variation.

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6. Product Performance Analysis

Top-performing products by revenue:

Product	Revenue
Chair	195,620.11
Printer	195,612.61
Laptop	192,126.56
Tablet	186,568.95
Monitor	175,651.41

Findings

- Chair and Printer generated nearly equal revenue.
- Electronics and office-related products dominate sales performance.

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7. Order Status Analysis

Distribution of customer orders:

Order Status	Count
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Findings

- Cancellation and return volumes are relatively high.
- Further investigation into operational efficiency may improve fulfillment performance.

8. Outlier Detection

Outliers were identified using the Interquartile Range (IQR) technique.

Variable	Outliers
Quantity	0
Unit Price	0
Items in Cart	0
Total Price	8

Findings

Only Total Price showed outliers, suggesting a small number of unusually large transactions.

9. Key Observations

1. The dataset is sufficiently clean for analysis.
2. Average order value equals **1,053.97**.
3. Revenue peaks indicate seasonal opportunities.
4. Product categories show clear performance differences.
5. High-value transactions impact average sales values.
6. Cancellation and return rates require additional investigation.

10. Conclusion

The Exploratory Data Analysis successfully identified important trends, customer purchasing behavior, and revenue drivers. Statistical analysis showed that sales are moderately variable and influenced by several high-value purchases.

The findings provide a strong foundation for deeper analysis and business decision-making.

11. Recommendations

- Develop dashboards for continuous monitoring.
- Investigate returned and cancelled orders.
- Perform customer segmentation analysis.
- Introduce predictive analysis for future sales forecasting.
- Visualize trends using Power BI or Tableau.

END OF REPORT